

Lesson 17 Relations

In mathematics, we have seen all kinds of *relations*, for example, *equality* of two numbers: $x = y$, and the “*less than or equal to*” relation: $x \leq y$. Notice that a relation relates *two* objects, as in the relation “ $\leq$ ” on a set $A = \{1, 2, 3\}$,

$$1 \leq 1, \quad 1 \leq 2, \quad 1 \leq 3, \quad 2 \leq 2, \quad 2 \leq 3, \quad 3 \leq 3$$

We may let R be the “ $\leq$ ” relation on A and write the relation as a set of *pairs*:

$$R = \{(1,1), (1,2), (1,3), (2,2), (2,3), (3,3)\} \subseteq A \times A$$

Recall the Cartesian product $A \times B = \{(a, b) : a \in A, b \in B\}$, consisting of *pairs*. So it is natural to use Cartesian product to define a relation relating objects in set A to objects in set B . This is a new way of looking at a familiar concept.

Definition Let A and B be sets. R is a **relation from A to B** if $R \subseteq A \times B$.

If $A = B$, then R is a **relation on A** and $R \subseteq A \times A$.

We use $(x, y) \in R$ and xRy *interchangeably* to mean that “ x is related to y ”. Be familiar with both notations! Similarly, we use $(x, y) \notin R$ and $x \not R y$ for “ x is NOT related to y ”. **A relation is a set of ordered pairs.**

Consider another example. Let R be the equality relation on $\mathbb{Z}$, then R a subset of $\mathbb{Z} \times \mathbb{Z}$, i.e. a set of ordered pairs (x, y) , where $x = y$.

$$R = \{ \dots, (-2, -2), (-1, -1), (0, 0), (1, 1), (2, 2), (3, 3), \dots \}$$

$$R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x = y\}$$

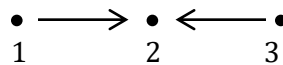
We say $(3, 3) \in R$ or $3R3$, meaning $3 = 3$, and $(1, 2) \notin R$ or $1R \not 2$, meaning $1 \neq 2$.

We may represent a relation **graphically** with arrows and dots. Let $A = \{1, 2, 3\}$.

a) $R = \{(1, 1), (2, 2), (3, 3)\}$ is a relation on A . Note that R is “equality” on A .

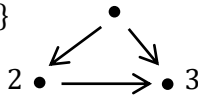


b) $S = \{(1, 2), (3, 2)\}$ is a relation on A .



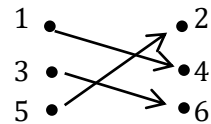
c) $T = \{(2, 3), (3, 4)\}$ is NOT a relation on A . Why? Because $(3, 4) \notin A \times A$.

d) $V = \{(x, y) \in A \times A : x < y\}$ is a “less than” relation on A : $V = \{(1, 2), (1, 3), (2, 3)\}$



Let $A = \{1, 3, 5\}$ and $B = \{2, 4, 6\}$. Then

a) $T = \{(1, 4), (3, 6), (5, 2)\}$ is a relation.



b) $S = \{(1, 1), (2, 2)\}$ is not a relation because $(1, 1) \notin A \times B$

Definition The **inverse** of a relation R is,

$$R^{-1} = \{(x, y) : (y, x) \in R\}$$

We have $(x, y) \in R^{-1}$ if and only if $(y, x) \in R$, reversing the arrows. For example, if $R = \{(1, 1), (2, 2), (3, 3)\}$, $R^{-1} = R$. If $S = \{(1, 2), (2, 3)\}$, $S^{-1} = \{(2, 1), (3, 2)\}$, and for $V = \{(x, y) \in A \times A : x < y\}$, $V^{-1} = \{(x, y) \in A \times A : x > y\}$.

Definition (Properties of Relations) Let R be a relation on set A .

- a) R is **reflexive** if xRx for all $x \in A$.

➤ There is a loop at every dot



- b) R is **symmetric** if xRy implies yRx for all $x, y \in A$.

➤ Between any 2 dots there are no arrows or arrows in both directions.



- c) R is **transitive** if xRy and yRz imply xRz for all $x, y, z \in A$.

➤ If 2 dots connect through another dot, then there is a direct arrow.



We will give a few examples.

- a) Let R be the equality relation “=” on $\mathbb{Z}$:

- R is reflexive, because $x = x$ for all $x \in \mathbb{Z}$
- R is symmetric, because for all $x, y \in \mathbb{Z}$, if $x = y$, then $y = x$.
- R is transitive, because for all $x, y, z \in \mathbb{Z}$, if $x = y$ and $y = z$, then $x = z$.

- b) Let R be the “less than or equal to” relation, i.e. “ $\leq$ ” on $\mathbb{Z}$,

$$R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x \leq y\}$$

- R is reflexive, because $x \leq x$ for every $x \in \mathbb{Z}$
- R is NOT symmetric, because $1 \leq 2$ but $2 \not\leq 1$.
- R is transitive, because for all $x, y, z \in \mathbb{Z}$, if $x \leq y$ and $y \leq z$, then $x \leq z$.

- c) Let R be the “less-than” relation, i.e. “ $<$ ” on $\mathbb{Z}$,

- R is NOT reflexive, because $x \not< x$ for any $x \in \mathbb{Z}$
- R is NOT symmetric, because $1 < 2$ but $2 \not< 1$.
- R is transitive, because for all $x, y, z \in \mathbb{Z}$, if $x < y$ and $y < z$, then $x < z$.

- d) Let A be a non-empty set, and R be the “is a subset of” relation, i.e. “ $\subseteq$ ”, on the subsets of A ,

$$R = \{(x, y) \in \mathcal{P}(A) \times \mathcal{P}(A) : x \subseteq y\}$$

- R is reflexive, because $B \subseteq B$ for every subset B of A .
- R is NOT symmetric: $\emptyset \subseteq A$ but $A \not\subseteq \emptyset$.
- R is transitive: Let X, Y , and Z be subsets of A . We have proved in Lesson 9 Exercises that if $X \subseteq Y$ and $Y \subseteq Z$, then $X \subseteq Z$.

- e) Let R be the “divides” relation on $\mathbb{Z}$

$$R = \{(x, y) \in \mathbb{Z} \times \mathbb{Z} : x|y\}$$

- R is reflexive, because $x|x$ for all $x \in \mathbb{Z}$.
- R is NOT symmetric, because $2|4$ but $4 \nmid 2$.
- R is transitive, because for all $x, y, z \in \mathbb{Z}$, if $x|y$ and $y|z$, then $x|z$.